

BODY - SIDE VIEW

from the right · scale 0.94 : 1 · dashed = hidden · purple = mechanism

HEAD = WINDING KEY
turn it 28x clockwise
(seen from above)

NECK RATCHET
12 teeth + flexure
pawl: winds one way

RUBBER MOTOR
3 loops of #16 band
= 6 strands x 60 mm

BODY: black back
shell + white belly
cap (2 snap-fit parts)

2 M8 STEEL WASHERS
low at the back
→ CG low + behind

TAIL + Ø8 ROLLER
= 3rd contact

PINION 12 T + 2 CRANKS (2.5 mm) on a Ø2
D-shaft under the crown's RIGHT rim
grease damper on its far end (A ⑥)

BIOMIMICRY



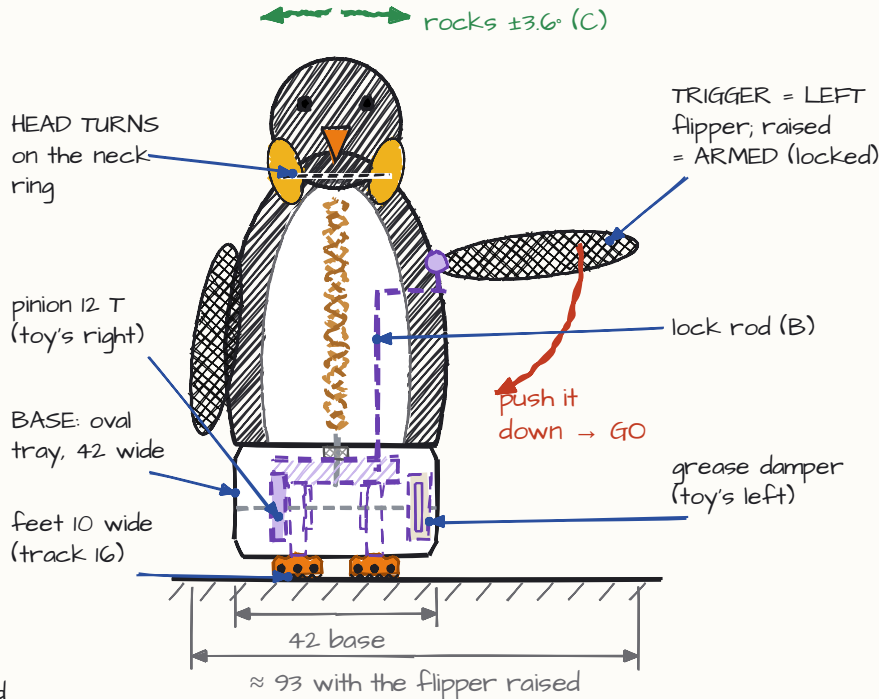
Emperors shuffle: short, low steps,
claws gripping the ice, body rocking
onto the stance foot, tail as a prop.
Copied: flat feet that never lift, claw
fins, lean onto the pushing foot, tail
roller. Bonus: the head swivels (it is
the winding key).

HOW IT PLAYS

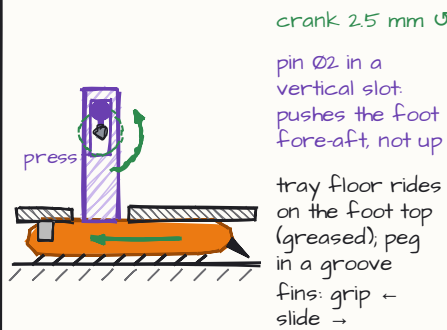
- ① Raise the LEFT flipper - click: the
lock rod springs onto the crown.
- ② Turn the head 28x clockwise
(click-click...), stop beak-forward.
- ③ Set it down on its feet and tail,
fully wound.
- ④ Steady its head with a finger, press
the flipper down, let go → it shuffles
and waddles for ≈ 30 s.

vs Concept 1: upright WALKER; TWISTED
rubber + crown lock pin; flipper trigger.

FRONT VIEW (armed) scale 0.64 : 1

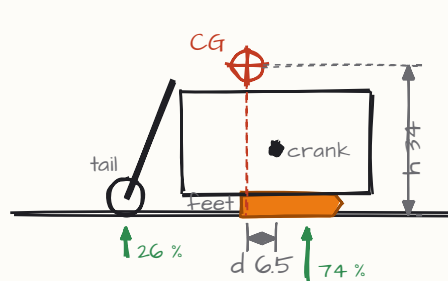


C FEET - SHUFFLE + WADDLE



Feet 180° apart: the back-going foot
grips ($\mu \approx 0.8$) and pushes the body
5 mm; the other skids on ($\mu \approx 0.15$):
10 mm per crank turn (≈ 9 with slip).
Waddle: at the bottom the pin presses
the sliding foot 1 mm down → body
rolls 3.6° onto the pusher (64% load).

D STANCE: 2 FEET + TAIL



Stands on both feet + its stiff tail
(Ø8 roller), always on 3 contacts.
Pushing-foot grip $0.8 \times 0.21 \text{ N} = 0.17 \text{ N}$
vs $\approx 0.02 \text{ N}$ needed → 8x margin.
Side tip needs 21° (feet span 26°, CG
34 high); waddle only $\pm 3.6^\circ$ ✓
Soft start (damper): no lurch, no nose-
dive. Steady the head when firing.

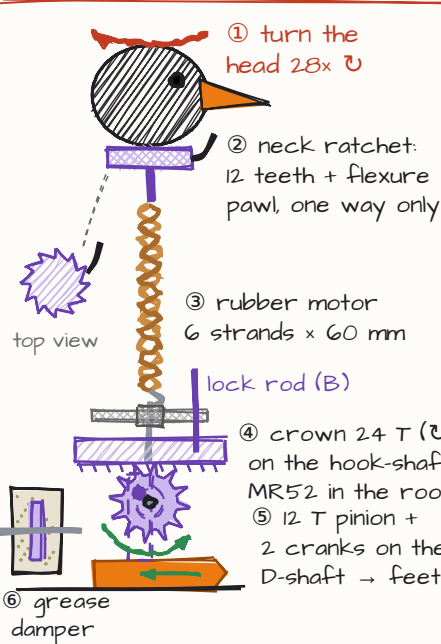
NUMBERS

- Mass $\approx 45 \text{ g}$, $\approx 86\%$ printed PLA
- CG 34 mm high, 6.5 mm behind the
crankshaft (74% on the feet)
- Motor $\approx 200 \text{ mJ}$; 5 s uses $\approx 52 \text{ mJ}$
- Gears ml, printed: crown 24 T →
pinion 12 T (x2), 2.5 mm cranks
- First 5 s: ≈ 8 strides $\approx 7 \text{ cm}$, walks
 $\approx 30 \text{ s}$ in all (≈ 40 strides, 0.36 m)
- Lock-rod side load $\leq 0.25 \text{ N}$
(2.3 Nmm ÷ 9.2 mm)
- Bought (10): D-shaft, tail axle,
hook-shaft, MR52, 3 bands, 2
washers, damping grease

RULES CHECK

- ✓ $\geq 75\%$ printed PLA ($\approx 86\%$)
- ✓ All parts fit the egg (box E);
the built toy is bigger
- ✓ Holds its energy until the
flipper is pushed down
- ✓ Trigger is a flipper crank - no
removable pin, stays together
- ✓ No energy stored when apart
(rubber untwisted)
- ✓ Snap / press fits, printed gears:
no glue, tools or toy parts
- ✓ Walks non-stop > 5 s (≥ 1
stride/s for 13 s, then slows)

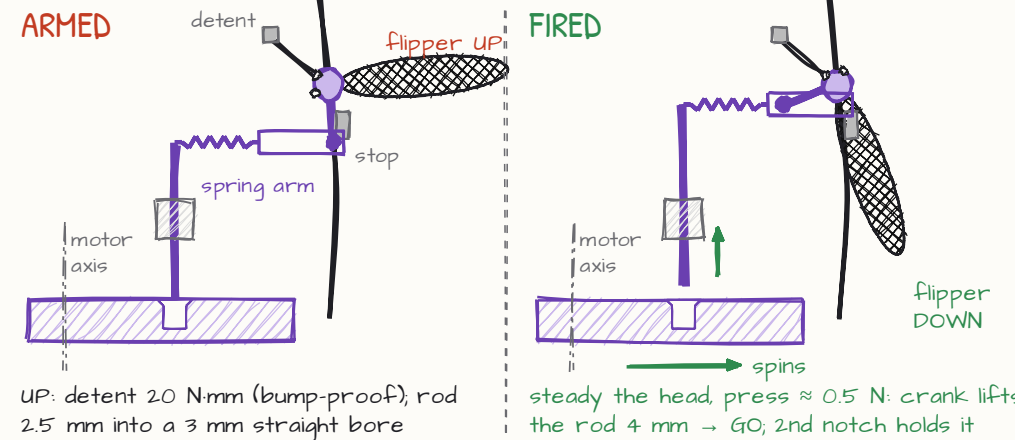
A ENERGY - TURN THE HEAD



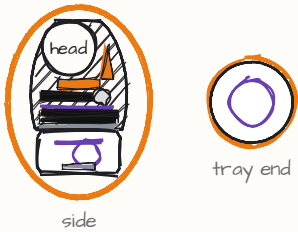
- Rubber: 3 loops of #16 band = 6 strands
x 60 mm (0.51 g), hook to hook
- 28 turns ($\approx 40\%$ of the ≈ 70 -turn break
point) → $400 \text{ J/kg} \times 0.51 \text{ g} \approx 200 \text{ mJ}$;
torque 2.3 N·mm, falls with turns left
- Crown 24 : pinion 12 → 2 crank turns per
motor turn. Crank torque $0.7 \times 2.3 \div 2$
 $= 0.81 \text{ Nmm}$; walking needs ≈ 0.22
(fins, tray rub, waddle lift), peak 0.35
- Undamped, the crank would race and
empty the motor in $\approx 1 \text{ s}$. ⑥ Damper: Ø10
rotor in 30 Pas silicone grease, 11 mm
gap each face → $c \approx 0.054 \text{ Nmm/s}$
- Cadence = $(\tau - \text{load}) \div 2\pi c$: 1.7 strides/s
at the start, 1.4 at 5 s, $\geq 1/s$ for 13 s,
 $\geq 0.5/s$ for 29 s (≈ 40 strides, 0.36 m)
- Worst cases (250 J/kg, load x2, damper
x2 or x2) each still walk $\geq 18 \text{ s}$. Weak
rubber AND rough floor: wind 40 turns

head → ratchet → rubber → crown → pinion (x2) → cranks → feet; ⑥ damper paces it

B TRIGGER - FLIPPER CRANK LOCK



E PACKS INTO THE EGG



Scale 1 : 4. The 2 body parts close into a hull
(56 tall with the open neck, 46 x 50) holding the
head, beak, feet, tail + roller, lock rod, flippers,
D-shaft and bands; the tray (crown + hook-shaft,
pinion, cranks, damper, MR52, washers) at the bottom.
Tightest: tray 21.4 of 23.3 mm; hull top 3.5 mm clear.
Assembled 109 mm tall → bigger than the egg ✓

ENGG*2100 F26 · Design & Build · Concept Sketch

Concept 2 · sheet 1 of 1

Concept 2: EMPEROR WADDLER

Theme: Biomimicry - emperor shuffle + waddle
Toy type: Walking toy (walks $\geq 5 \text{ s}$ non-stop)
Scale / units: mm · side view 0.94 : 1 on 11x17

Drawn by: _____ Signature: _____ Date: _____